

1. Show that: $\vec{e}_i = g_{ij}\vec{e}^j$.
2. Under a coordinate transformation q^i to q'^j , show that the basis vectors transform as

$$\vec{e}'_j = \frac{\partial q^i}{\partial q'^j} \vec{e}_i \quad (1)$$

3. Given the definition of the Riemann tensor

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma} \quad (2)$$

Show that $R^\rho_{\sigma\mu\nu}$ transforms as a tensor of rank 4.

4. $R_{ijkl} = g_{ia} R^a_{jkl}$. Prove that:

$$R_{ijkl} = -R_{jikl} = -R_{ijlk} = R_{klij} \quad (3)$$

5. Prove the Bianchi identity

$$R_{ijkl} + R_{iklj} + R_{iljk} = 0 \quad (4)$$

6. (a) Compute the Riemann tensor for a 2-sphere of radius R .
 (b) Compute the Riemann tensor for the surface of a right circular cone with half opening angle α .

7. Prove the following identities:

$$\begin{aligned} g_{ij} g^{jk}{}_{,l} &= -g^{jk} g_{ij,l} \\ g^{ij}{}_{,l} &= -\Gamma^i_{kl} g^{kj} - \Gamma^j_{kl} g^{ki} \\ |g|_{,i} &= -|g| g_{kj} g^{kj}{}_{,i} = |g| g^{kj} g_{kj,i} \end{aligned}$$

where $|g|$ is the determinant of g_{ij} .

$$g_{ij,k} = \frac{\partial}{\partial x^k} g_{ij} \quad (5)$$

8. Are the Christoffel symbols a mixed tensor?

9. If F_{ik} is an antisymmetric tensor then show that

$$Z_{ikl} \equiv \frac{\partial F_{ik}}{\partial x^l} + \frac{\partial F_{kl}}{\partial x^i} + \frac{\partial F_{li}}{\partial x^k}$$

is a third rank tensor.

10. Self study: Study from 'A first course in GR by B. Schutz' an alternative approach of the derivation of Riemann Tensor. He has showed that by computing the change in the vector by parallel transporting itself in an infinitesimal closed loop on the curved manifold, one can obtain the Riemann curvature tensor.